From Farm and Weather Data to Solar and Battery Trading Decisions

Data, ensemble formulation and real-time delivery

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Abstract

A solar farm owner must commit electricity before its production is known. Battery storage provides flexibility, while successive auctions create opportunities to revise the contracted position as weather information improves. We study this problem using four northern-Italian farms, historical production and irradiance, regularly updated weather forecasts, and 50-member weather ensembles. Weather is converted into possible AC production paths that are used to plan physically feasible trading and battery decisions. Auction participation studies investigate the importance of jointly optimizing current trades and future adjustments. This report explains the available data, the forecast-to-decision process, the deterministic and ensemble formulations, and physical delivery control. The full optimization and supporting equations are collected in appendices.

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1 The problem and the core idea

A photovoltaic (PV) farm and a battery share a grid connection. Solar generation can be exported, stored or curtailed; the battery can be charged by the local farm or by grid imports while also being able to discharge to the network. There is no local load. The producer sells and buys energy in the Italian day-ahead market (MGP) and the three intraday auctions (MI1, MI2 and MI3). Later auctions can adjust the resulting position, while the battery has the double function of separating instantenous generation from future trades and of offering flexibility for real time operations when production differs from forecasts.

The core idea is to jointly optimize at each auction the current market positions and expected future adjustments given available information and present (or projected) State of Charge (SoC). This allows to take advantage of price spreads while respecting physical constraints of our assets, speculating on market prices up until delivery. Only current auction trades are binding while future ones can still be updated as newer production forecasts are available. The actual battery state links the decisions across delivery intervals and days.

We formulate the problem as a Mixed-integer Linear Program (MILP) and solve it to optimality via GUROBI. The model maximizes expected remaining auction revenue less battery wear and trading costs, subject to energy conservation, exclusive (binary) charging/discharging, stored-energy bounds and the shared grid limit. The full optimization is in Appendix A.

We assess profits together with imbalance, curtailment, wear and ending stored energy, while assuming to be a price-taker market participant.

2 The available data

2.1 Four farms and their historical records

The dataset covers four anonymous farms in the northern-Italian bidding zone. The supplied metadata describe fixed arrays and single-axis trackers and the battery are hypothetical study assets:

Farm	AC capacity (MW)	Declared technology	Battery (MWh)	Power (MW)
1	0.95328	Fixed array	0.75	0.5
345	1.52460	Fixed array	1.00	0.5
901	4.16000	Bifacial single-axis tracker	2.00	1.0
910	6.63800	Single-axis tracker	5.00	2.5

Their energy and power ratings, 7–93% state-of-charge range, 90% round-trip efficiency and connection limits are modeling assumptions. Exact farm coordinates are unavailable, so solar-position calculations use documented geographical proxies;

Historical records run from 1 January 2025 to 31 August 2026 on a 15-minute UTC grid: 58,368 timestamps for each farm. They contain AC power, historical *satellite* global horizontal irradiance (GHI), and historical diffuse fraction.

2.2 Updating deterministic weather forecasts

Given that market positions must be traded mostly the day before delivery, obtaining accurate PV production forecasts for the next day is imperative. We use weather-to-pv models to obtain such estimates.

The main weather dataset comes from the ECMWF (European Centre for Medium-range Weather Forecast) archive containing 4 deterministic weather forecasts cycles per day (initialized every six hours) from 30 December 2024 to 31 August 2026. Each run simulates up until 48 hours from the initialization time. The weather fields include several irradiance components, temperature, wind and cloud cover.

Several cycles predict the same delivery interval with newly available information. This provides a record of forecast revision as delivery approaches. The four daily issues are successive information updates, each supplying one deterministic weather trajectory.

In the studies, forecasts are assumed to become available six or seven hours after initialization.

2.3 Weather ensembles

Two separate archives supply *50 weather members for the same issue and lead times*, with the same four farms and 0–48-hour lead coordinates:

These members represent alternative weather trajectories within one forecast issue. The spread supplies uncertainty information that a single deterministic forecast cannot describe. As more data becomes available, the ensemble can provide a more comprehensive view of potential outcomes, and act as training data for developing more sophisticated forecasting generation methods for stochastic methods.

2.4 Market observations

Historical clearing curves cover DA and the three intraday auctions, with imbalance-settlement prices used to value actual delivery deviations. Physical operation uses quarter-hours intervals resulting in 96 time steps per day, mirroring the products granularity.

The participation comparisons use actual future auction prices as a common price-information benchmark. This isolates choices about market access, production forecasts and physical flexibility under that benchmark. Their profits must be interpreted with this price-foresight assumption. We are also developing price prediction models based on load and generation forecasts as well as past historical prices. The first results show that real operating profits with price proxies are around a third or a fourth compared to simulations using actual realized prices. Real bidding and auction behavior has not been included in this work yet.

3 From weather information to production and decision paths

3.1 Selecting information at the auction cutoff

The study uses the following market timeline, in Italian local time. Decisions are taken five minutes before each gate:

Auction	Gate	Delivery covered	Latest typical weather issue
DA	Previous day, 12:00	Whole next day	Previous day, 00 UTC
IDA1	Previous day, 15:00	Whole next day	Previous day, 06 UTC
IDA2	Previous day, 22:00	Whole next day	Previous day, 12 UTC
IDA3	Delivery day, 10:00	Noon to midnight	Delivery day, 00 UTC

The last column follows the six-hour release assumption and requires sufficient remaining-horizon coverage. In our data weather forecasts have a 48-hour horizon allowing us to have predictions for

trading in day ahead markets. At every cutoff, the deterministic forecast is updated to the latest eligible issue. Given the raw data size of all 4 cycles of the ensemble, only the 00 UTC is used for the ensemble studies.

3.2 Converting weather into AC generation

The baseline physical forecast is taken from the pvlib library. It first derives direct and diffuse irradiance components, then projects them onto the array or tracker plane. It estimates module temperature from air temperature, wind and plane-of-array irradiance, computes DC production and converts it to AC power. Output is bounded by the farm AC rating and forced to zero during the specified seasonal nighttime windows. Effective conversion parameters can be estimated from historical data.

Statistical and time-series neural models provide another mapping from weather, calendar, lead-time and farm information to AC production. Historical power and irradiance support fitting and diagnosis; evaluation uses the forecast weather available for the delivery interval. Where learning uses historical irradiance, the shift to forecast irradiance is stated explicitly. Chronological training, selection and evaluation periods are kept distinct. Measured-only and reconstructed-target scores answer different questions.

Each ensemble weather member passes separately through the selected conversion model, producing a complete AC power path on the delivery grid. The study's 51-path input combines these 50 paths with the updating deterministic forecast. This allows us to generate a set of possible coherent scenarios instead of using independent sampling. Future studies, when more data becomes available, will investigate ways to generate similar temporally correlated trajectories for a more statistically sound stochastic optimization.

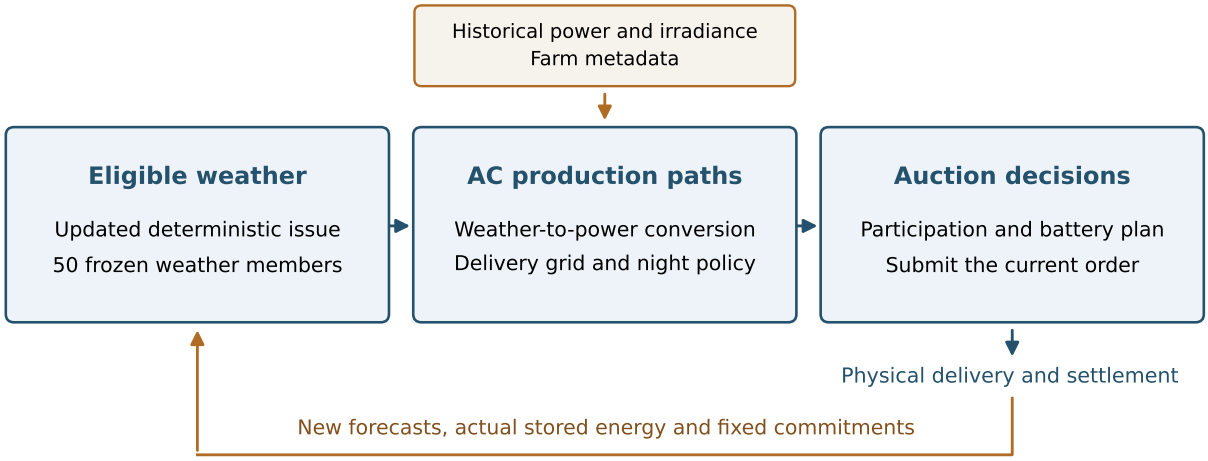


Figure 1: Historical records support the production mapping and evaluation. Eligible weather issues supply forecast paths; those paths support auction and physical decisions. Realized delivery supplies the carried battery state.

3.3 Turning forecast paths into a trading policy

At a given cutoff, a production path is a possible external input. A decision path consists of auction quantities and feasible battery/solar actions along the represented information history. Scenarios with the same information must share the same trajectory.

3.3.1 Deterministic and ensemble formulations

The deterministic case optimizes against one eligible AC production forecast. It chooses auction quantities, charging/discharging and curtailment so that the forecast grid exchange exactly matches the cumulative contracted position in each delivery interval. Energy, power, grid and terminal constraints apply to this one trajectory. At a later auction the same problem is solved again with updated information and the battery state appropriate to that decision.

The ensemble case uses 50 production paths from the preceding delivery day's 00 UTC weather issue plus the updating deterministic path, each weighted equally at 1/51. Auction quantities and the entire planned battery schedule are shared across all represented paths. Their common initial energy and charging/discharging imply a common stored-energy trajectory. Solar curtailment can differ with available production. For contracted energy Q_t, common battery powers c_t, d_t and path-specific curtailment $u_{s,t}$, exact delivery requires

$$(V_{s,t} - u_{s,t} + d_t - c_t)\Delta_t = Q_t, \qquad 0 \le u_{s,t} \le V_{s,t}, \qquad \forall s, t. \tag{1}$$

Thus one submitted position and battery plan must be feasible under every represented production path. Other possibilities like allowing each path to have its own schedule while sharing the first action is a possible extension.

	Deterministic case	Strict ensemble case
Production input	One updating forecast	50 frozen weather paths and one updating forecast
Trade and battery plan	One schedule	One schedule shared across all paths
PV curtailment	For the single forecast	Adapts to production in each path
Planned delivery	Exact for one forecast	Exact for every represented path
Main effect	Feasibility for the forecast trajectory	Feasibility over the intersection of pathwise constraints

Under the common actual-price curves used here, exact delivery makes useful solar injection identical across paths. It is therefore limited by the lowest available production at each interval: $0 \le i_t \le \min_s V_{s,t}$, whereas the deterministic case has $0 \le i_t \le V_{0,t}$. This envelope is an exact simplification under the shared-control assumptions.

The objective retains a scenario-weighted expectation, but weights do not soften constraints imposed on every member. This explains the conservatism of this strict formulation. Allowing planned deviations or future battery actions conditional on new observations defines a different stochastic formulation and is being investigated.

3.4 Reoptimizing the remaining decisions

Integrated planning optimizes the remaining participating auctions together. Only the closing auction's quantity is submitted. At the next gate, that order is fixed (or in future iterations updated with actual accepted trades) and provisional later orders are reoptimized with new eligible forecasts. During delivery, commitment tracking adjusts physical battery power within its energy, power and grid limits. Actual SOC, submitted commitments, wear and transaction-fee state carry into subsequent decisions and days. A forecast projection supplies the starting state for tomorrow's plan while today's delivery is still in progress; the realized state governs execution.

4 Imbalances and real-time control

Even an exactly balanced forecast plan can produce a realized imbalance: actual exported energy differs from the submitted position, and the signed deviation is valued at the applicable settlement price. For realtime operation, a feedback policy must be chosen to minimize imbalances. Our baseline is a *greedy grid-tracking policy*. In each quarter-hour, it adjusts battery power to bring grid exchange toward the contracted target, discharging for a shortfall and charging for a surplus within available power and SOC limits. It releases scheduled curtailment when useful and redirects otherwise-curtailed PV into the battery when charging power and SOC headroom permit, while preserving the grid-exchange target. Remaining excess PV is curtailed. This local correction uses interval-average PV feedback in the replay; it does not optimize the consequences for later intervals. The current *model predictive control (MPC)* study instead reoptimizes over the remaining delivery day every 15 minutes, using the current battery state, fixed commitments and eligible deterministic or ensemble PV forecasts. It trades expected imbalance volume or forecast settlement value against wear and terminal energy, executes only the first action and repeats. The comparison asks whether anticipating later shortfalls and preserving flexibility improves delivery and operating profit relative to greedy tracking. The physical controller submits no new auction orders, but the changed battery state affects subsequent auction decisions.

Table 1: Continuous greedy grid-tracking results oracle prices, 1 January–31 August 2026.

Metric	Farm 1 (new)	Farm 1 (previous)	Farm 345	Farm 901	Farm 910
Total installed capacity (MW)	0.95328	0.95328	1.52460	4.16000	6.63800
Forecasting Model	BiGRU	BiGRU	BiGRU	TCN	TCN
Realized operating profit (EUR)	165,806.38	164,206.15	271,676.98	967,777.75	1,547,926.25
Auction revenue (EUR)	181,953.78	176,836.03	298,667.43	1,058,105.79	1,747,432.44
Signed imbalance settlement (EUR)	-9,832.26	-5,932.07	-16,724.55	-68,157.65	-154,668.67
Degradation cost (EUR)	6,315.14	6,697.81	10,265.91	22,170.39	44,837.52
Gross PV production before curtailment (MWh)	684.70	684.70	1,325.14	5,358.40	8,367.90
Absolute imbalance (MWh)	56.78	36.71	97.41	412.48	926.08
Absolute imbalance / gross PV (%)	8.29	5.36	7.35	7.70	11.07
Signed net imbalance (MWh)	-54.73	-33.44	-93.80	-410.41	-918.67
PV curtailment (MWh)	21.78	43.54	227.02	204.53	225.70
PV curtailment / gross PV (%)	3.18	6.36	17.13	3.82	2.70
Battery charge (MWh)	120.27	127.56	195.46	422.22	853.02
Battery discharge (MWh)	108.25	114.81	176.02	380.04	769.46
Final state of charge (%)	49.01	48.90	38.87	47.77	13.20
Retained metered AC (intervals)	14,828	14,828	14,491	14,885	10,212
Monthly night zero (intervals)	8,332	8,332	8,332	8,332	8,332
Reviewed pvlib replacement (intervals)	137	137	40	72	4,713
Retained-exclusion pvlib (intervals)	27	27	459	26	67

Notes: All runs use the latest available deterministic forecast cycle, greedy grid tracking and actual auction-price curves in planning. Gross PV is available AC production before curtailment and may contain synthetic values to isolate outages. Percentages use gross PV over the entire replay as their denominator. Signed imbalance settlement is realized cash flow.

5 Participation studies: the importance of joint optimization

The four auctions give 15 nonempty participation profiles. We compare their integrated planning with sequential all-auction planning, which optimizes only the current auction's trade, and a perfect-production reference. Each profile carries its own SOC, commitments, wear and fees chronologically. The January 2026 comparison starts at 90% SOC and uses a 50% planning terminal target. Figure 2 summarizes farm 1. Actual future auction prices are shared across paths, so integrated planning also benefits from anticipating cross-auction spreads.

For farm 1's all-auction profile, ensemble planning reduces absolute imbalance from 5.818 to 1.577 MWh, while realized curtailment increases from zero to 0.775 MWh relative to deterministic

Table 2: Continuous greedy grid-tracking results with a daily 50% auction-planning terminal SOC, 1 January–29 August 2026.

Metric	Farm 1	Farm 345	Farm 901	Farm 910
Total installed PV capacity (MW)	0.95328	1.52460	4.16000	6.63800
Forecasting model	BiGRU	BiGRU	TCN	TCN
Realized operating profit (EUR)	163,338.71	263,162.85	952,546.10	1,524,282.09
Auction revenue (EUR)	173,432.03	280,831.45	1,022,908.66	1,667,040.39
Signed imbalance settlement (EUR)	-3,514.05	-7,187.83	-48,457.27	-95,521.57
Degradation cost (EUR)	6,579.27	10,480.78	21,905.29	47,236.72
Gross PV production before curtailment (MWh)	677.35	1,310.74	5,302.05	8,279.71
Absolute imbalance (MWh)	22.35	46.12	307.70	614.51
Absolute imbalance / gross PV (%)	3.30	3.52	5.80	7.42
Signed net imbalance (MWh)	-20.49	-40.52	-302.19	-600.62
PV curtailment (MWh)	24.33	278.97	241.71	272.00
PV curtailment / gross PV (%)	3.59	21.28	4.56	3.29
Battery charge (MWh)	125.29	199.50	416.89	898.56
Battery discharge (MWh)	112.78	179.76	375.77	810.74
Final state of charge (%)	47.65	28.07	19.67	7.00
Retained metered AC (intervals)	14,696	14,359	14,753	10,212
Monthly night zero (intervals)	8,272	8,272	8,272	8,272
Reviewed pvlib replacement (intervals)	137	40	72	4,581
Retained-exclusion pvlib (intervals)	27	459	26	67
User-requested zero (intervals)	0	2	9	0

Notes: The 50% terminal SOC is imposed in the auction plan for each delivery day; the greedy real-time controller does not enforce it. The final SOC reported is the actual end-of-replay SOC. Runs use the latest eligible deterministic forecast cycle and actual auction prices in planning. PV model selection and current-interval realized PV are retrospective. Gross PV is available AC production before curtailment and includes curated synthetic values and reconstructed exclusions. Percentages use gross PV over the replay as the denominator. Signed imbalance settlement is realized cash flow; negative net imbalance indicates a shortfall. Degradation is a modeled cost, and profit does not value final battery inventory. Night zeros and synthetic values are not independent metered observations.

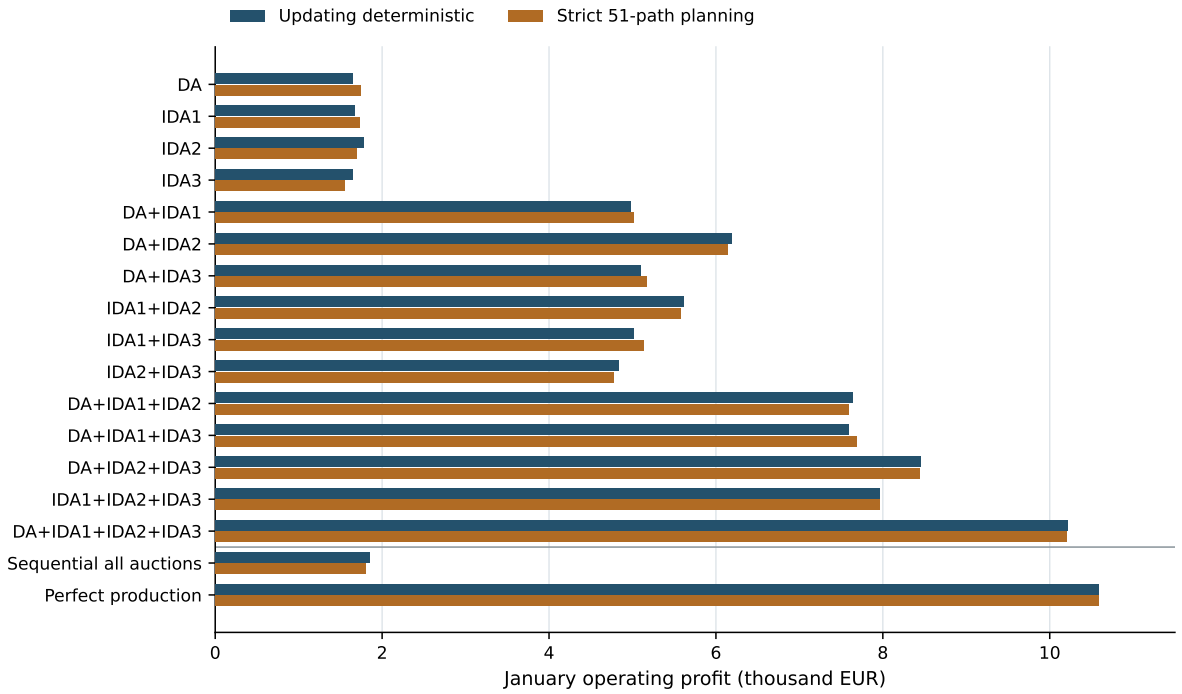


Figure 2: Farm 1, continuous January 2026: monthly operating profit for all 15 auction subsets and the two references. Paired bars compare the updating deterministic and strict 51-path formulations under actual-price foresight. DA = Day Ahead, IDA=IntraDay Auction

planning. This illustrates the trade-off between tighter delivery tracking and curtailed solar energy. These curtailment totals are the residual after real-time feedback, which can redirect otherwise-curtailed PV into the battery when charging power and SOC headroom are available.

Future studies will repeat this experiment with predicted prices, measuring how uncertainties interact with profits in all-auction participation.

6 What this framework enables next

This work connects forecast revision, production uncertainty and sequential participation within one physical chronology. The next questions are how forecast calibration and price prediction change those participation results, and how receding-horizon battery control uses new observations between auction gates. These developments build on the same data-to-path-to-decision structure. Our framework combines multiple ideas from the literature like multi-auction participation and scenario-based stochastic optimization with accurate and historically available weather data and realtime feedback controllers.

A Full residual auction optimization

A.1 Sets, parameters and signs

Let $t \in \mathcal{T}$ index the remaining physical intervals of a delivery day, Δ_t their duration in hours, $s \in \mathcal{S}$ scenarios with probabilities w_s, and $a \in \mathcal{A}_k$ auctions still open at decision k. $\mathcal{T}_a$ is auction a's eligible delivery set. Native products group physical intervals; their identifiers retain daylight-saving offsets.

Solar availability is $V_{s,t}$ MW AC, battery capacity E MWh, power limit $\bar{P}$ MW, energy bounds $E^{\min}, E^{\max}$, efficiencies η_c, η_d and hourly self-discharge σ. The grid limits are $P^{\text{imp}}, P^{\text{exp}}$. Q_t^{fix} is the signed MWh position from all earlier submitted quantities, including still-unpriced orders. Positive trade/grid signs mean sale/export. Auction price coefficients are $\pi_{a,s,t}$ in EUR/MWh.

Controls $c_{n,t}, d_{n,t}$ are charging/discharging MW; $z_{n,t}$ is a charging binary. Curtailment is $u_{m,t}$ MW, injection $i_{s,t}$ MW, grid exchange $g_{s,t}$ MW and internal energy $e_{s,t}$ MWh. $q_{a,n,t}$ is a signed auction adjustment in MWh. Node maps $n_t(s), n_a(s), m_t(s)$ encode observable history. Leaf values below refer to their node-owned variables, so indistinguishable histories cannot take different actions.

A.2 Complete base problem

For throughput wear $D_s = c_{\text{thr}} \sum_t (c_{s,t} + d_{s,t})\Delta_t$ and optional import cost $I_s = f \sum_t [-g_{s,t}]_+ \Delta_t$, the residual model is

$$\max \quad \sum_s w_s \left[\sum_{a \in \mathcal{A}_k} \sum_{t \in \mathcal{T}_a} \pi_{a,s,t} q_{a,n_a(s),t} - D_s - I_s - C_s^{\text{trade}} \right] \tag{2}$$

$$\text{s.t.} \quad i_{s,t} + u_{m_t(s),t} = V_{s,t}, \quad i_{s,t} \geq 0, \quad u_{m_t(s),t} \geq 0, \qquad \forall s,t, \tag{3}$$

$$g_{s,t} = i_{s,t} + d_{n_t(s),t} - c_{n_t(s),t}, \qquad \forall s,t, \tag{4}$$

$$-P^{\text{imp}} \leq g_{s,t} \leq P^{\text{exp}}, \qquad \forall s,t, \tag{5}$$

$$0 \leq c_{n,t} \leq \bar{P} z_{n,t}, \quad 0 \leq d_{n,t} \leq \bar{P}(1 - z_{n,t}), \quad z_{n,t} \in \{0,1\}, \qquad \forall n,t, \tag{6}$$

$$e_{s,t+1} = (1-\sigma)^{\Delta_t} e_{s,t} + \eta_c c_{n_t(s),t} \Delta_t - d_{n_t(s),t} \Delta_t / \eta_d, \qquad \forall s,t, \tag{7}$$

$$E^{\min} \leq e_{s,t} \leq E^{\max}, \quad e_{s,0} = e^k, \qquad \forall s,t, \tag{8}$$

$$Q_{s,t} = Q_t^{\text{fix}} + \sum_{a : t \in \mathcal{T}_a} q_{a,n_a(s),t}, \qquad \forall s,t, \tag{9}$$

$$g_{s,t} \Delta_t = Q_{s,t}, \qquad \forall s,t, \tag{10}$$

$$-P^{\text{imp}} \Delta_t \leq q_{a,n,t} \leq P^{\text{exp}} \Delta_t, \qquad \forall a,n,t \in \mathcal{T}_a, \tag{11}$$

$$-P^{\text{imp}} \Delta_t \leq Q_t^{\text{fix}} + \sum_{b \leq a : t \in \mathcal{T}_b} q_{b,n_b(s),t} \leq P^{\text{exp}} \Delta_t, \qquad \forall a,s,t, \tag{12}$$

$$q_{a,n,t}/\Delta_t = q_{a,n,t'}/\Delta_{t'}, \qquad t, t' \text{ in the same native product.} \tag{13}$$

Only eligible quantities exist. The last equality ties average power within a native product; on equal quarter-hours it ties MWh shares directly. Fixed initial energy can be observed or a forecast projection to the eligible boundary. A configured terminal equality $e_{s,T} = e_F$ and/or floor $e_{s,T} \geq e_{\text{floor}}$ is added where the study requires it. A planning terminal target does not itself force an algebraic feedback controller to end at that realized energy.

Earlier auction cash and incurred wear are sunk and absent from (2). Only the closing auction's order is submitted; other quantities are revisable plans. Sequential planning replaces $\mathcal{A}_k$ by the single current auction.

B Battery wear, control and realized accounting

B.1 Wear models

For replacement value R, reference internal cycle depth d_0, life N and capacity E, AC throughput wear can be calibrated as

$$c_{\text{thr}} = \frac{R}{N d_0 E(1/\eta_c + \eta_d)}, \qquad D = c_{\text{thr}} \sum_t (c_t + d_t)\Delta_t. \tag{14}$$

An optional segmented depth model uses stress $\Phi(d) = \alpha d^\beta$, $\beta \geq 1$, J segments and marginal discharge cost

$$m_j = \frac{RJ}{\eta_d E}\{\Phi(j/J) - \Phi((j-1)/J)\}, \qquad D = \sum_{t,j} m_j d_{j,t} \Delta_t. \tag{15}$$

Segment energy transitions sum to aggregate energy; shallow-first charging and discharging priorities preserve cycle-depth memory. The segment vector carries between decisions and cannot generally be recovered from SOC alone. This is an approximation to cycle wear [1].

A current study tries to integrate a relatively simple calendar and cycling aging of the battery, while considering in the optimization problem the resulting shrinking capacity updating at the end of the delivery day.

For replacement value R, nominal capacity E_0, reference cycle depth d_0, cycle life N, and committed state of health h_d on delivery day d, the available capacity is $E_d = E_0 h_d$. Calendar and cycling capacity losses are fractions of nominal capacity and accumulate as

$$\Delta q_{\text{cal}} = \gamma_{\text{cal}} \frac{\Delta t}{T_{\text{yr}}}, \qquad \Delta q_{\text{cyc}} = \frac{\Delta E_{\text{int}}}{2 N d_0 E_0}, \qquad h_{d+1} = 1 - q_{\text{cal},d+1} - q_{\text{cyc},d+1}, \tag{16}$$

where γ_{cal} is the calendar-loss fraction per year and $T_{\text{yr}} = 365 \times 24$ hours, $\Delta E_{\text{int}} = (\eta_c c + d/\eta_d)\Delta t$ is internal charge-plus- discharge energy, and c and d are AC charge and discharge powers. The optimizer uses the current committed SOH for that delivery day, forecasts losses for future capacity constraints, and commits the updated SOH only at the end of each chronological day. Forecast damage is priced once as

$$C_{\text{wear}} = \frac{R}{1 - h_{\text{EOL}}} \left(\Delta q_{\text{cal}} + \Delta q_{\text{cyc}}\right), \tag{17}$$

where h_{EOL} is the end-of-life SOH threshold. This linear cycling approximation depends on internal throughput, not cycle depth. For the study calibration, $\gamma_{\text{cal}} = 0.01\,\text{yr}^{-1}$, $N = 6000$, $d_0 = 1$, and $h_{\text{EOL}} = 0.8$; the factor of two converts charge-plus-discharge energy to equivalent full cycles.

B.2 Feedback, chronology and profit

Commitment tracking uses target $g_t^* = Q_t^{\text{sub}}/\Delta_t$ and interval PV. Write $\rho_t = (1-\sigma)^{\Delta_t}$ for energy retention. Feasible charge and discharge are bounded by

$$C_t = \min\left(\bar{P}, \frac{[E^{\max} - \rho_t e_t]_+}{\eta_c \Delta_t}\right), \quad D_t = \min\left(\bar{P}, \frac{\eta_d \left[\rho_t e_t - E^{\min}\right]_+}{\Delta_t}\right). \tag{18}$$

It adjusts net battery power toward the grid target, recovers curtailed PV where headroom permits, and enforces connection limits. Interval-average PV feedback is an idealized information assumption. Executed battery state, curtailment and deviations are retained at every interval.

Realized accounting uses

$$\delta_t = g_t^{\text{real}} \Delta_t - Q_t^{\text{sub}}, \tag{19}$$

$$J^{\text{real}} = \sum_{a,t} \pi_{a,t}^{\text{real}} q_{a,t}^{\text{sub}} + \sum_t \left(\lambda_t^+ [\delta_t]_+ - \lambda_t^- [-\delta_t]_+ \right) - D^{\text{real}} - I^{\text{real}} - C^{\text{trade,real}}. \tag{20}$$

With a single settlement price this reduces to $\lambda_t \delta_t$. Forecast penalties, tail-risk terms and terminal continuation values are absent from realized cash. Final energy is reported separately. The state and binding positions carry across days; no daily physical reset occurs in a continuous comparison.

C Forecast construction and information timing

C.1 Issue, availability and delivery

For forecast initialization τ_v, lead h and availability A_v, define

$$t_{v,h} = \tau_v + h, \qquad A_v = \tau_v + \ell_{\text{release}}, \tag{21}$$

$$v^*(k) = \max\{v : A_v \le k, \text{ required delivery horizon covered}\}. \tag{22}$$

Here ℓ_{release} is the study's assumed six- or seven-hour release lag; the main-body auction table illustrates the six-hour case. The deterministic production forecast at cutoff k is $V_{0,t}^{(k)} = f_\theta(X_{v^*(k),t}, M)$, where M denotes farm assumptions. For delivery day d, let v_d be the preceding day's 00 UTC ensemble issue. The path input is

$$V_{s,t}^{(k)} = f_\theta(X_{v_d,s,t}, M), \quad s = 1, \dots, 50, \qquad w_s = \frac{1}{51}, \quad s = 0, \dots, 50. \tag{23}$$

The deterministic path updates while the weather-ensemble issue is fixed for the day. Model availability and observation delays are additional eligibility conditions; retrospective fitted models retain their historical-use label.

C.2 Physical and statistical production mappings

For GHI, horizontal beam irradiance BHI and solar zenith θ_z, the physical conversion uses the symbolic relationships implemented in the pvlib library:

$$G_{\text{DHI}} = G_{\text{GHI}} - G_{\text{BHI}}, \qquad G_{\text{DNI}} \simeq G_{\text{BHI}} / \cos\theta_z, \tag{24}$$

$$G_{\text{POA}} = G_{\text{beam}} + G_{\text{sky}} + G_{\text{ground}}, \tag{25}$$

$$T_{\text{mod}} = T_{\text{air}} + G_{\text{POA}} / (U_0 + U_1 v), \tag{26}$$

$$P_{\text{DC}} = \max\left(0, \frac{P_{\text{AC,rated}}}{\eta_{\text{ref}}} \frac{G_{\text{POA}}}{1000} [1 + \gamma(T_{\text{mod}} - T_{\text{ref}})]\right), \tag{27}$$

$$V_t = \mathbf{1}_{\{t \text{ outside the seasonal night window}\}} \operatorname{clip}\big(\mathcal{C}(P_{\text{DC}}), 0, P_{\text{AC,rated}}\big). \tag{28}$$

Low-sun handling and valid irradiance bounds are required for the DNI conversion. $\mathcal{C}$ represents the selected DC-to-AC/output-loss mapping. The reference coefficients are $U_0 = 25$, $U_1 = 6.84$, $\gamma = -0.004/^\circ\text{C}$ and $\eta_{\text{ref}} = 0.96$; fitted historical conversions retain their own parameters. Tracker position and any rear contribution use explicit geometry assumptions.

A statistical model instead learns parametrized (with parameters θ) mappings from inputs to the normalized target $y_t = P_t / P_{\text{AC,rated}}$:

$$\hat{\theta} = \arg\min_\theta \frac{1}{|\mathcal{D}_{\text{train}}|} \sum_{t \in \mathcal{D}_{train}} \rho\big(f_\theta(X)_t - y_t\big) \tag{29}$$

Squared or Huber loss fits point predictions; normalization and preprocessing are learned from training data. Calendar, weather and lead/issue features must be available at the cutoff. A sequence model may use the whole eligible issued weather sequence, including future forecast covariates. Future observed production is reserved for evaluation or an explicit perfect-production reference.

For ensemble values x_s, probabilities w_s and realized power y, marginal quality can be measured by

$$\mathrm{CRPS} = \sum_s w_s |x_s - y| - \frac{1}{2} \sum_{s,r} w_s w_r |x_s - x_r|. \tag{30}$$

RMSE, MAE, bias and coverage provide complementary diagnostics. These scores evaluate forecasts separately from realized profit, physical feasibility and serial dependence. A calibrated collection of marginals requires an explicit temporal coupling before it defines new production paths for storage.

References

[1] B. Xu et al., *Factoring the Cycle Aging Cost of Batteries Participating in Electricity Markets*, 2017. https://arxiv.org/abs/1707.04567.